

Weather and Climate Basics

Weather is short-term atmospheric condition, while climate describes long-term patterns and averages.

Key facts:

Weather includes temperature, humidity, wind, precipitation, cloud cover, and atmospheric pressure.

Climate is commonly described using statistics over decades, often a 30-year reference period.

Air pressure differences help drive winds from higher pressure toward lower pressure areas.

Warm air can hold more water vapor than cold air, which affects cloud formation and precipitation.

Ocean currents transport heat and influence regional climates.

Greenhouse gases absorb and emit infrared radiation, helping regulate Earth's surface temperature.

Climate variability includes natural patterns such as seasonal cycles and ocean-atmosphere oscillations.

Answerable questions:

Q: What is the difference between weather and climate? A: Weather is short-term atmospheric condition; climate is the long-term pattern of weather statistics.

Q: Why do winds form? A: Winds form largely because pressure differences and Earth's rotation move air through the atmosphere.